



Local And Web Ui AI

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Purpose

The purpose of this lab is to download and run an AI locally on your personal machine and then get it up onto the network. Additionally, this Lab adds the complexity of creating multiple AIs through different ways specialized for different tasks.

Background

AI, also known as artificial intelligence, is a specialty within computer science that is concerned with creating systems that can replicate human intelligence and problem-solving abilities. They do this by taking in a myriad of data, processing it, and learning from their past to streamline and improve in the future. A normal computer program would need human interference to fix bugs and improve processes. It has been a thing since the early-to mid-2000s. Back then, it was primarily used as some basic algorithms that could perform basic tasks and basically predict what the user would do. These types of algorithms would include autocorrect, NPCs and characters in video games, search engines, and even seemingly advanced ones like AlphaGo.

The foundation of modern AI that we all know today was invented in 2017 when Google introduced the Transformer model, a neural language processing framework that would become the basis for GPT and virtually all contemporary large language models. Building on this architecture, OpenAI released GPT-1 in 2018, followed by GPT-2 in 2019, which surpassed the 1 billion parameter threshold. Most recently, GPT-5 released in 2025, representing the most sophisticated and capable version of the technology to date. GPT is used by millions of users worldwide.

Local AI began around 2023 and is AI that can be run on local devices. Most advanced models by then were exclusively available through cloud-based APIs, creating challenges for research working in isolated networks, organizations with strict data privacy requirements, and users seeking offline functionality. The ability to have an AI locally in your system solves those problems.

One of the first publicly available local AIs is LLaMA. Meta's LLaMa, released in February 2023 and intended exclusively for academic researches and developers, demonstrated that powerful language models could operate effectively on local systems without constant internet connectivity. When the model came out, the broader community recognized the potential for Local AI access and began developing different Local AI models.

Parallel to model development, the tools to simplify local AI deployments developed. Applications like LM Studio, Ollama, and Docker-based solutions emerged to make local AI accessible beyond command-line interfaces, providing GUI (Graphical user interfaces) and streamline installation processes to install models that lowered technical barriers to develop these models.

New Local AI models span a wide parameter range, from lightweight 1 billion parameter models suitable for basic tasks to 100 billion parameters requiring multi-GPU.



configurations. This lab focuses on models between 5 and 35 billion parameters, a range that balances capability with the computational power of the GPU in our PCs.

Docker is a containerization platform that packages applications and their dependencies into standardized units called containers. In the context of Local AI development, Docker provides crucial containers that ensure that AI models run consistently across different environments by bundling all necessary libraries, frameworks, and configurations together.

Ollama is a tool designed specifically for running large language models locally with minimal configuration. Released in 2023, it simplifies the process of downloading, managing, and running open-source AI models on personal computers. The platform supports a wide variety of open-source models including Llava:13b, mistral, phi, gemma, and many others. Ollama uses a command-line interface like Docker, with simple commands like `Ollama run llama3` to download and execute models.

LM studio is a desktop application that provides a graphical user interface for discovering, downloading, and running local language models as well. It offers a more user-friendly alternative to command-line tools, featuring a intuitive interface that appeals to users who prefer visual interaction over terminal commands. LM Studio provides advanced features such as adjustable inference parameters (temperature, top-p, context length), conversation history management, and the ability to test models with different prompting strategies. The software includes a local API server that mimics the OpenAI API format, allowing developers to build applications against a familiar interface while using local models.

Open WebUI is a self-hosted web interface designed to provide a ChatGPT-like experience for local AI models. It serves as the front-end layer that makes the local AI models accessible through browser-based chat interfaces that a user can access from any device. The platform offers a lot of customization options including multiple chat sessions, conversation history, model switching, and user management features. Open WebUI offers us a space where we can test our Local AIs on other devices not on our PCs.

Lab Summary

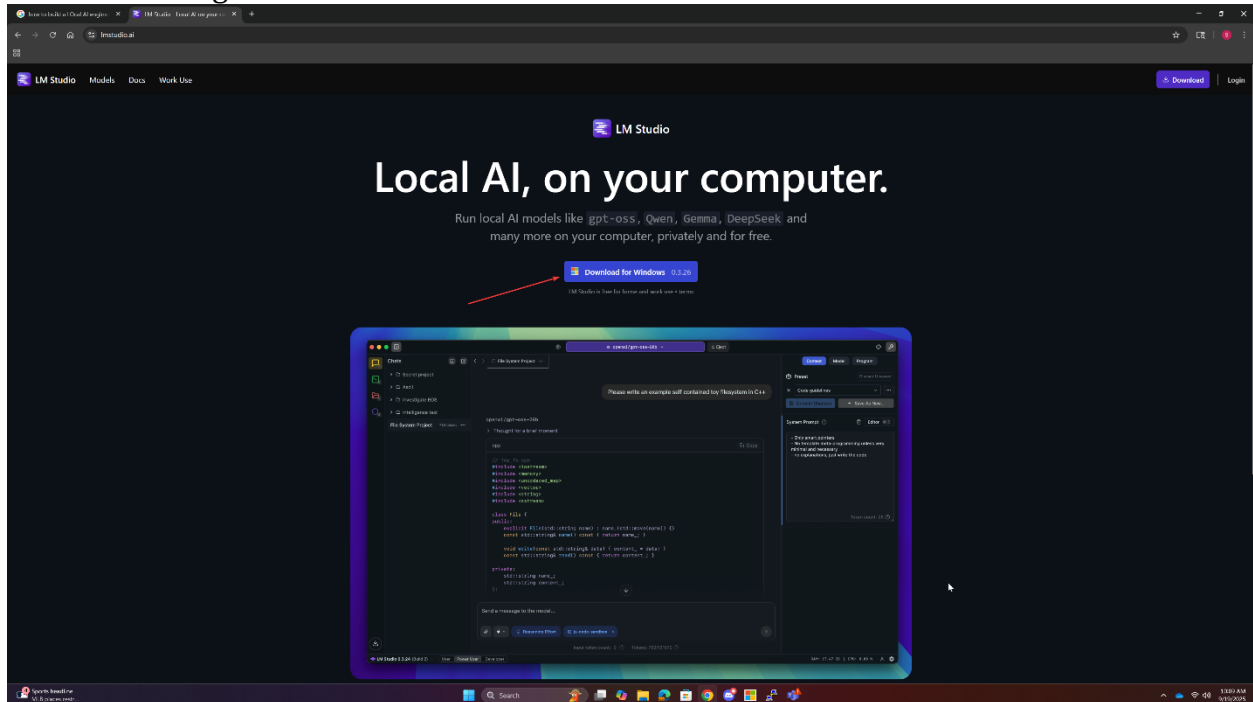
In this lab, I created two different Local AIs, one through LM studio and the other using Ollama. I installed LM studio, and downloaded and ran a AI from there. I also installed Docker Desktop, ran a basic local setup on Docker itself with basic local AI. I then also installed Ollama locally for the web UI setup. I then installed Open WebUI for Docker and then ran it on a browser via port 3000. I then set up an account for myself and went to install models from the Ollama website and started using it. I finally figured out the IP address of wlan0 on the PC, and then I used that IP to fish up the Web UI on another device that was on the same network.

Lab Procedure

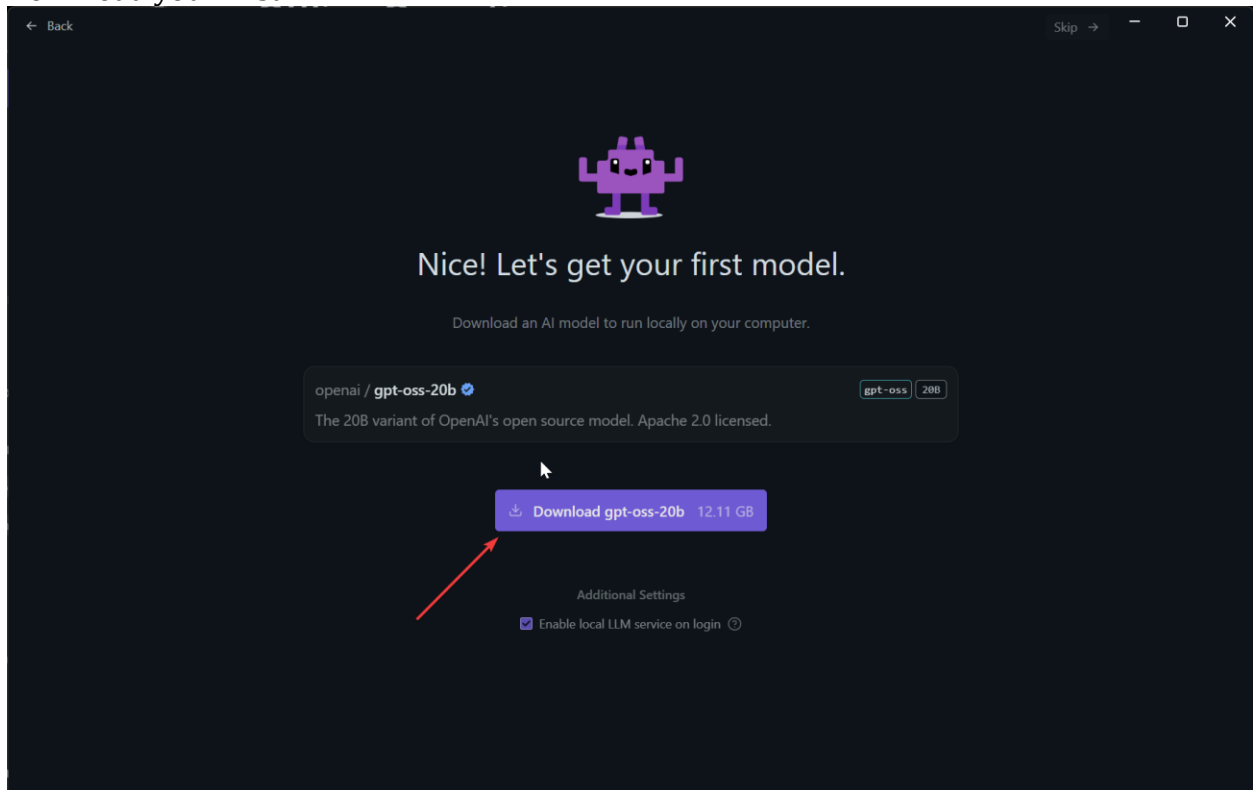
LM studio:



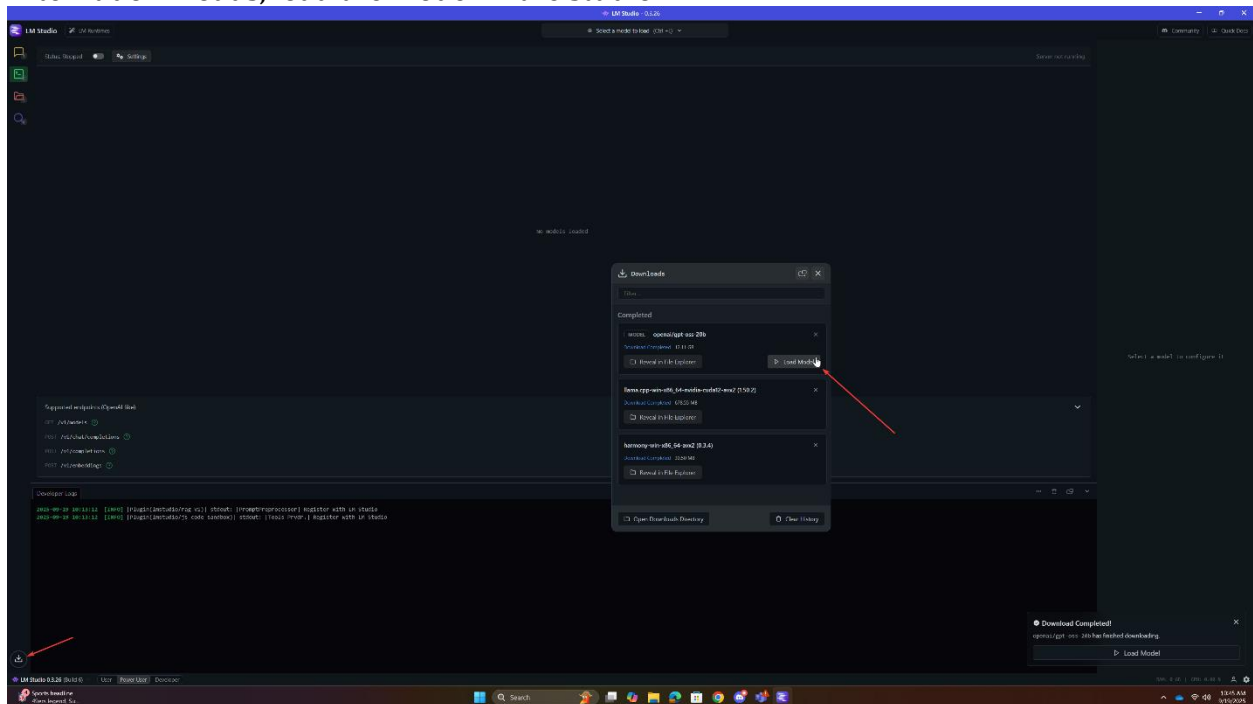
This first AI is really straightforward to install and get it running.
For the first AI go to [Lmstudio.ai](https://lmstudio.ai) and click download for windows:



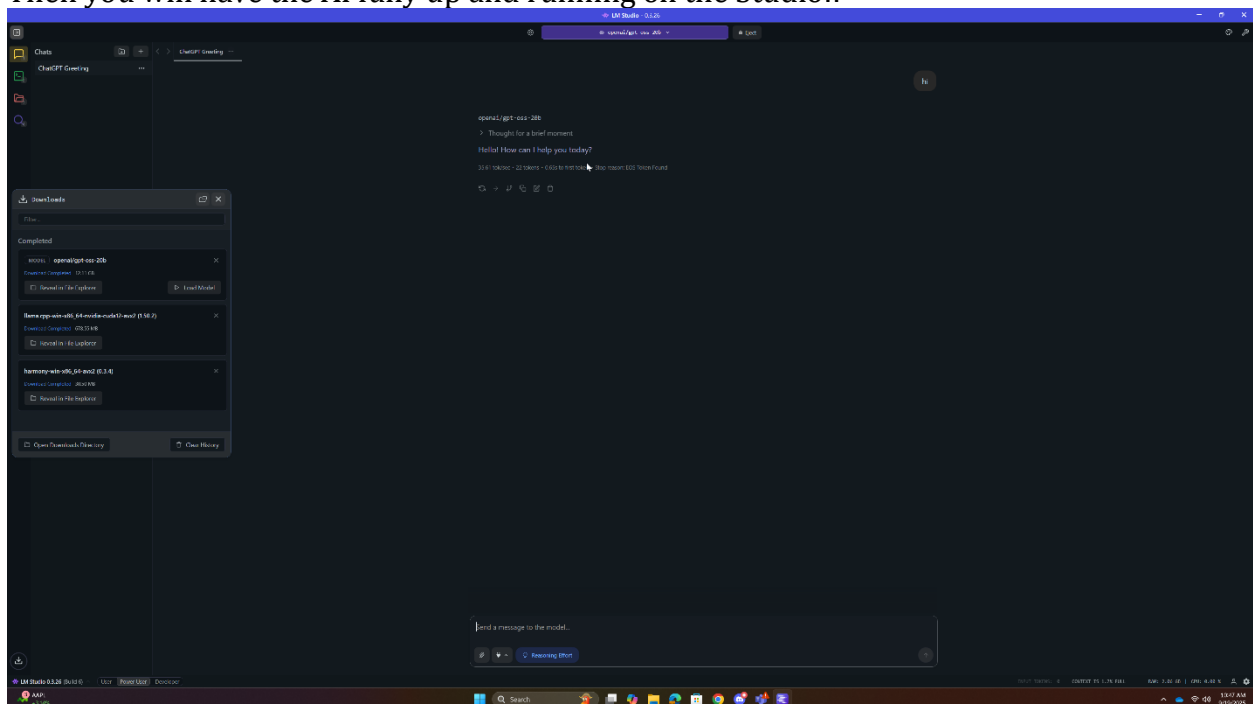
Download your first AI:



After it downloads, load the model in the studio:



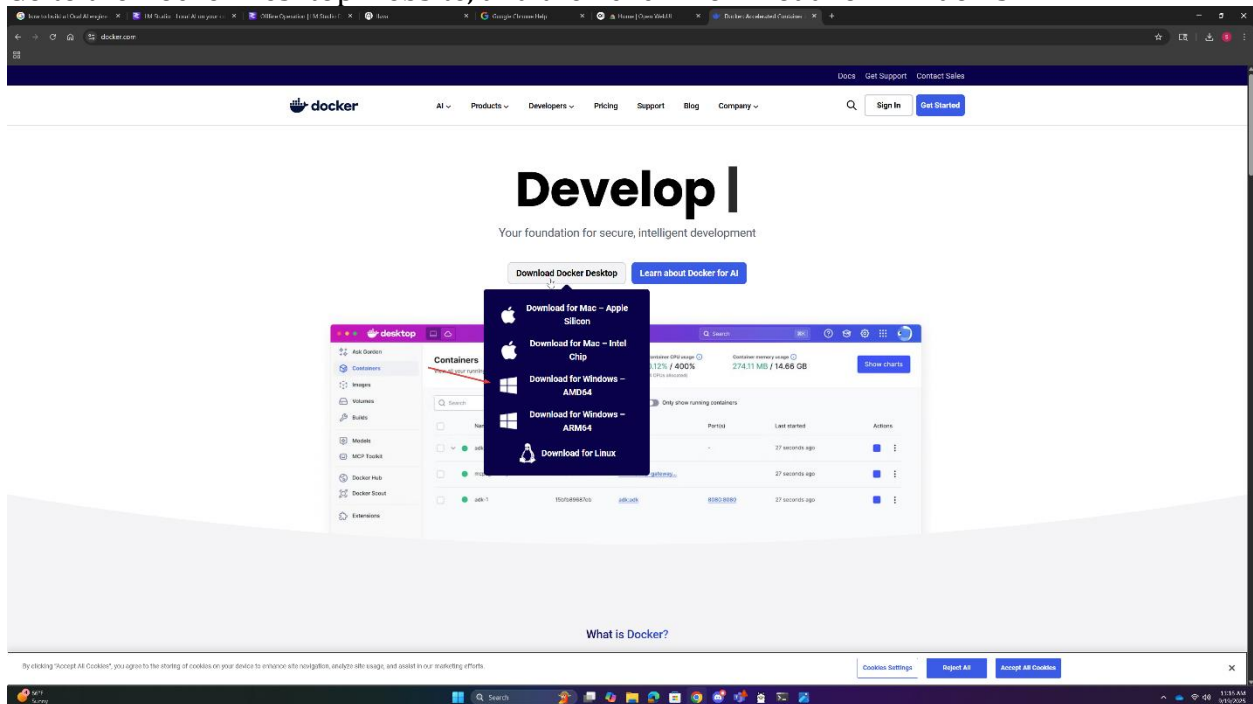
Then you will have the AI fully up and running on the Studio!:



Ollama



Go to the Docker Desktop website, and then click 'Download for Windows':

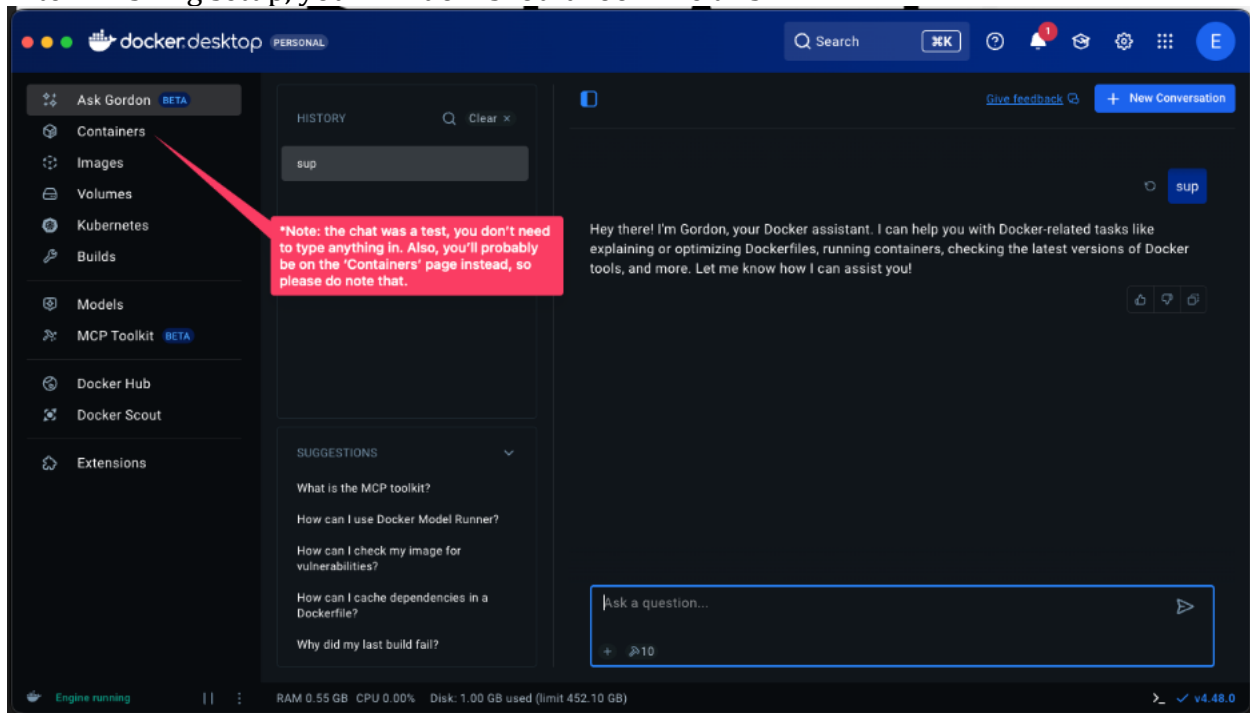


Afterwards, run the setup for Docker. It will prompt you to create an account—you do so in the browser—and then it may ask for you to install certain AI models.

The Setup process will also ask you if you're using Docker for Personal use or Enterprise use. In this case, just select 'PERSONAL' and continue.

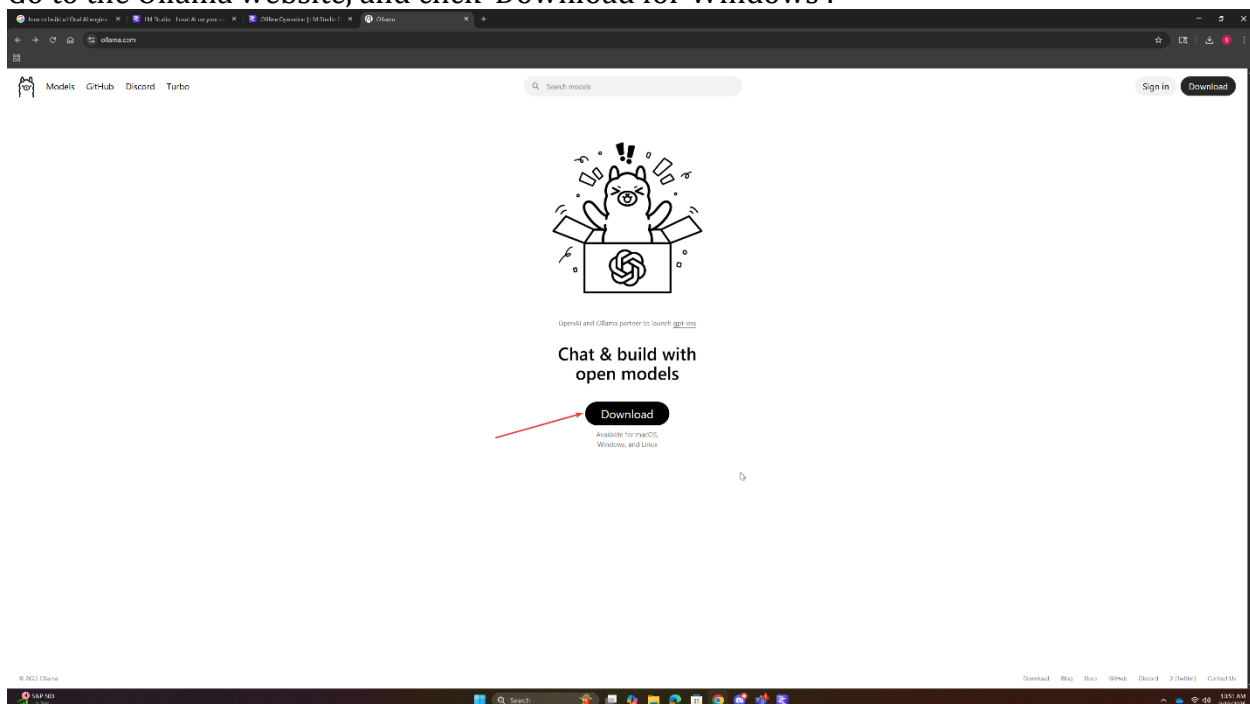


After finishing setup, your window should look like this



Install Ollama.

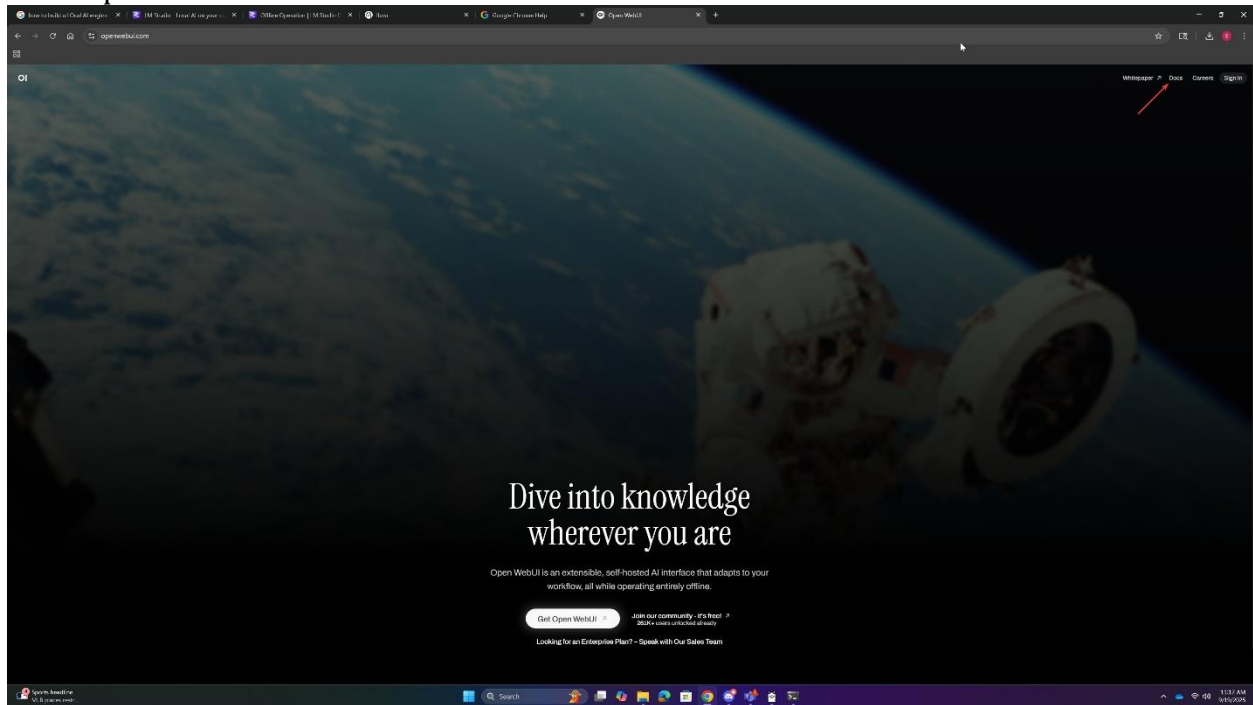
Go to the Ollama website, and click 'Download for Windows':



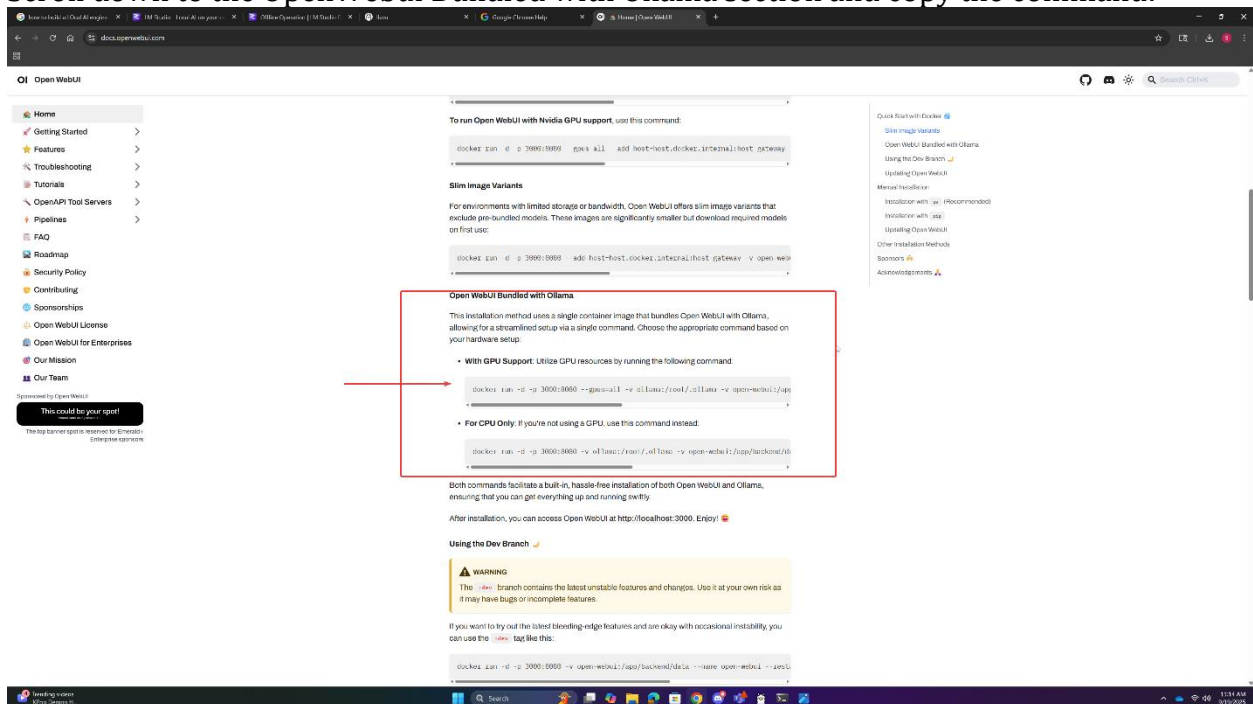
Finish the setup up and follow the instructions, and then run it.



Go to openwebui.com and click on the docs



Scroll down to the OpenWebui Bundled with Ollama section and copy the command:



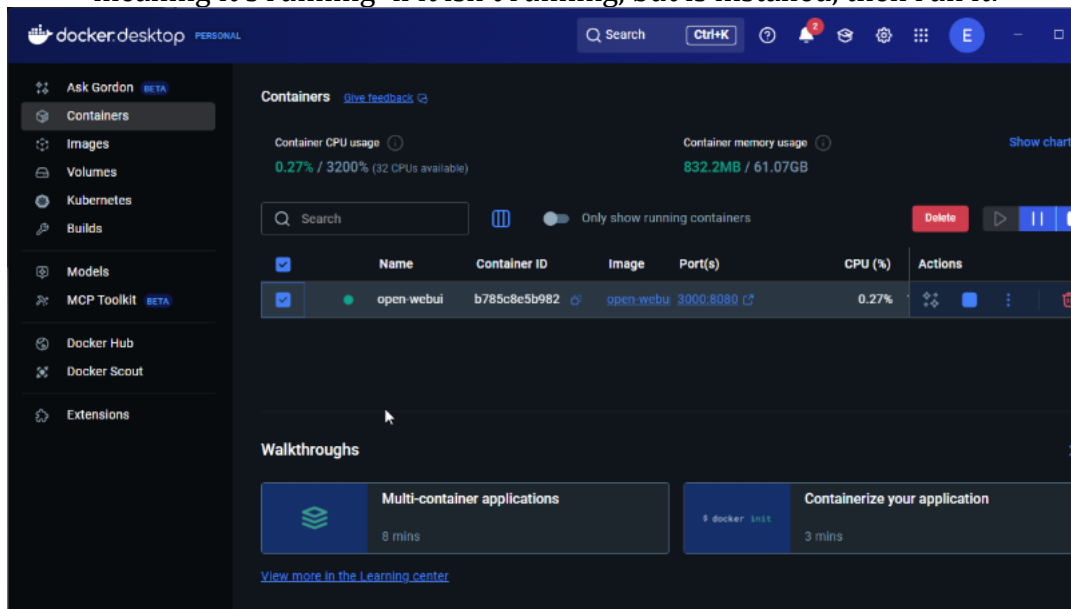
Go to the Command Prompt.

Now type in this code from the Open Web UI Docs

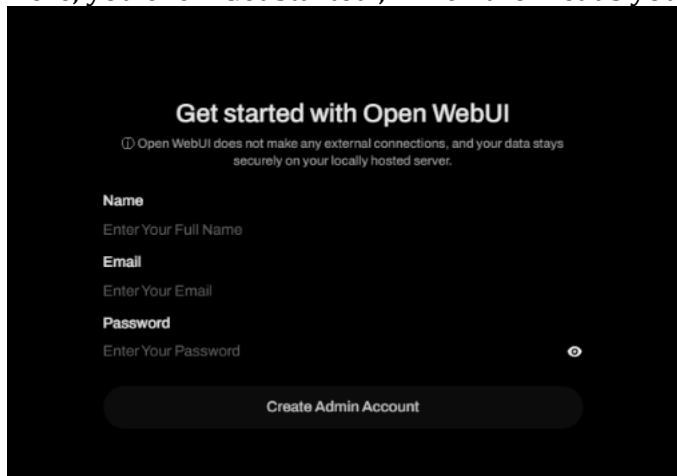
```
C:\Users\elysiaUwU>sudo docker run -d -p 3000:8080 --gpus=all ollama:/root/.ollama -v open-webui:/app/backend/data --name open-webui --restart always ghcr.io/open-webui/open-webui:ollama
```

```
Command Prompt - docker r X + v
pulling d5ca8c59f62d: 100% 46 B
pulling 17b7e63f6e77: 100% 51 B
pulling b15ee2b77419: 100% 490 B
verifying sha256 digest
writing manifest
success
>>>
>>>
C:\Users\efy1228>docker run -d -p 3000:8080 --gpus=all -v ollama:/root/.ollama -v open-webui:/app/backend/data --name o
en-webui --restart always ghcr.io/open-webui/open-webui:ollama
Unable to find image 'ghcr.io/open-webui/open-webui:ollama' locally
ollama: Pulling from open-webui/open-webui
659036234d55: Pulling fs layer
d107e437f729: Pull complete
4f4fb700ef54: Download complete
1fd1c0f46ae6: Download complete
ce4392d98604: Download complete
60b67f313695: Download complete
a830e79592b2: Download complete
f8043d35c69a: Download complete
4079926b32d5: Download complete
b0b6ac0f5373: Pulling fs layer
8740d48eeda5: Download complete
1ce95d9ae66c: Pulling fs layer
f1451649e055: Downloading [>] 5.243MB/1.258GB
7552d7e25e4a: Pulling fs layer
057d7f346e4f: Downloading [=====] 14.68MB/39.61MB
49b1c40da033: Downloading [>] 3.146MB/341.1MB
```

Afterwards, it should install, and if you go back to Docker Desktop, then you'll see this, meaning it's running—if it isn't running, but is installed, then run it:

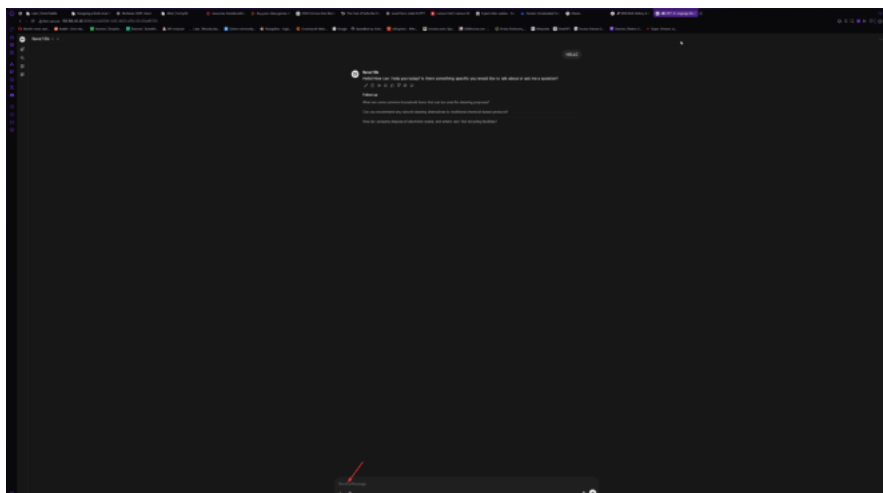
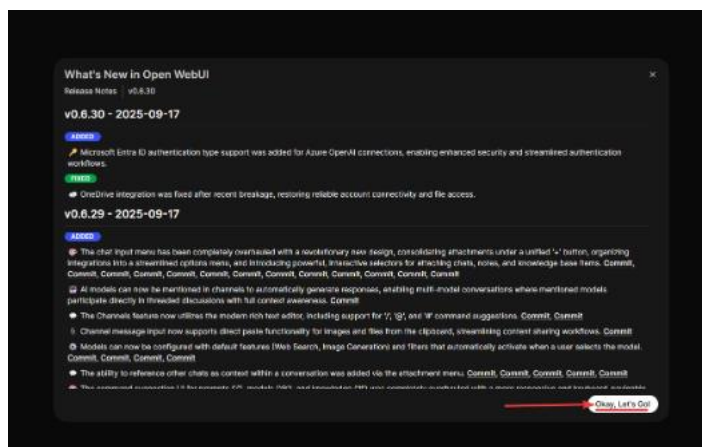


Here, you click 'Get started', which then leads you to the account creation page:

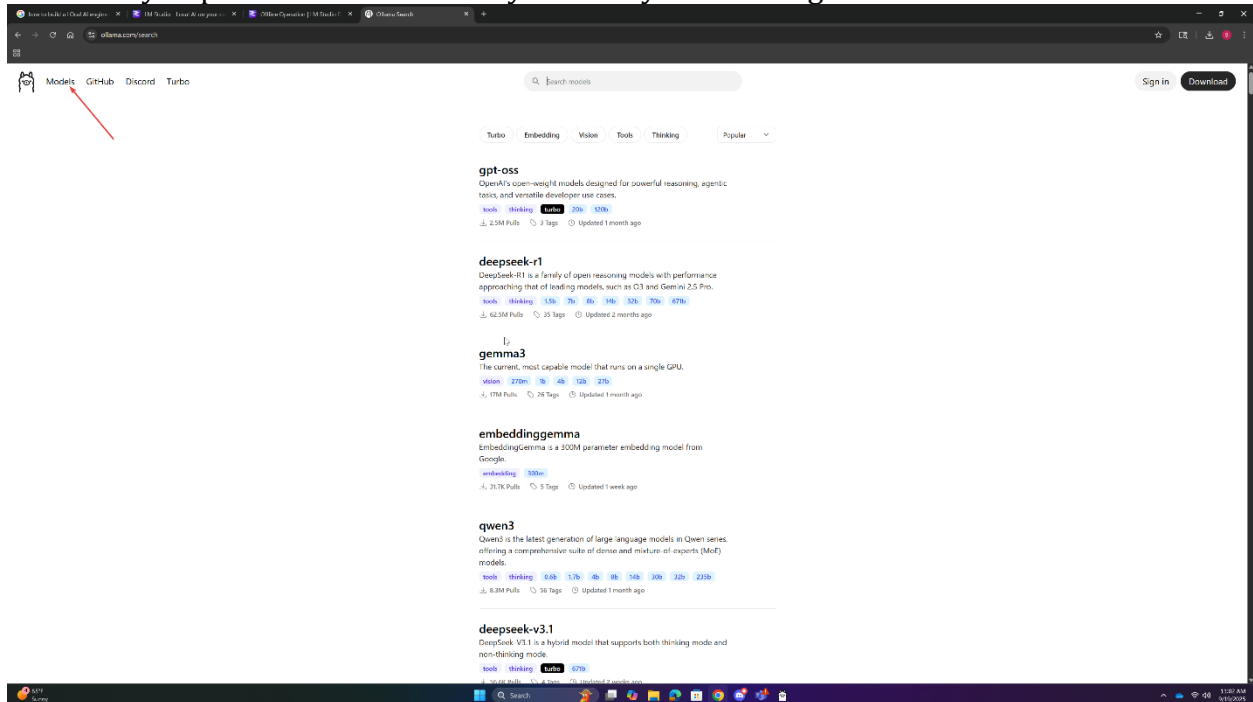


The screenshot shows a dark-themed web interface for creating an account. At the top, it says "Get started with Open WebUI". Below this, a note states: "Open WebUI does not make any external connections, and your data stays securely on your locally hosted server." The form consists of three input fields: "Name" with the placeholder "Enter Your Full Name", "Email" with "Enter Your Email", and "Password" with "Enter Your Password" and an eye icon for toggling visibility. At the bottom is a large "Create Admin Account" button.

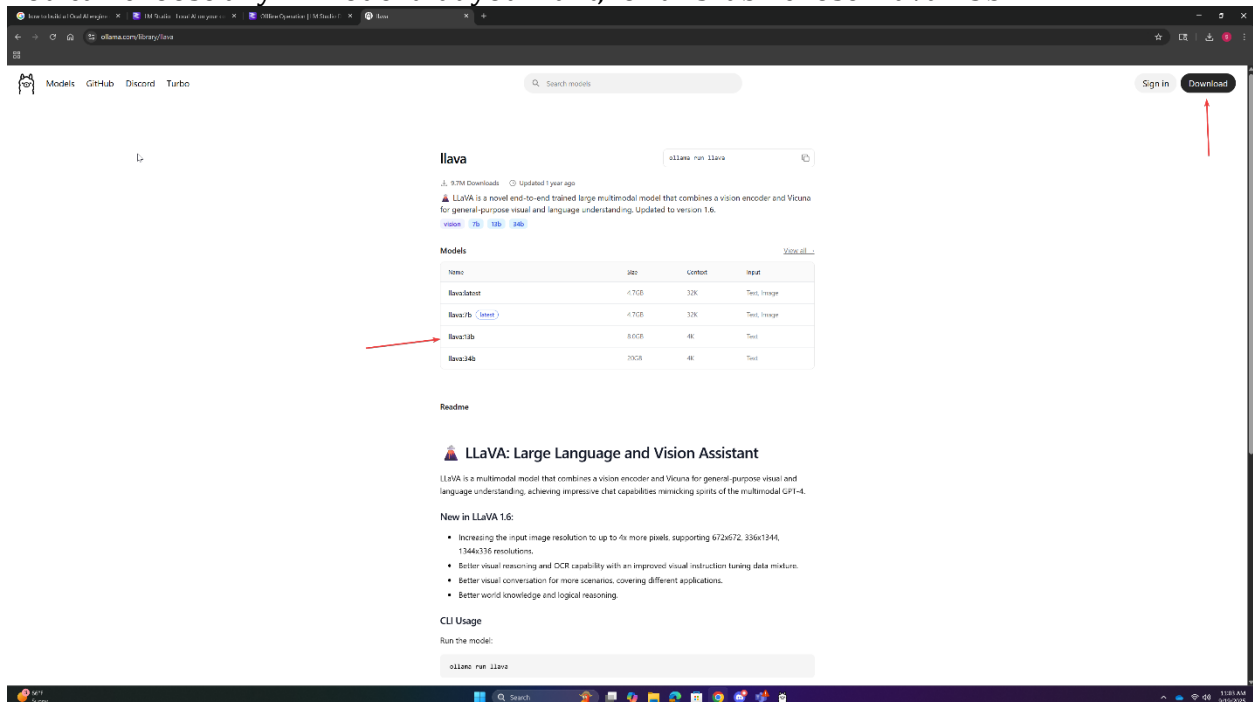
After you create an account, click on Okay Lets Go! and you'll end up on a screen that looks something like this



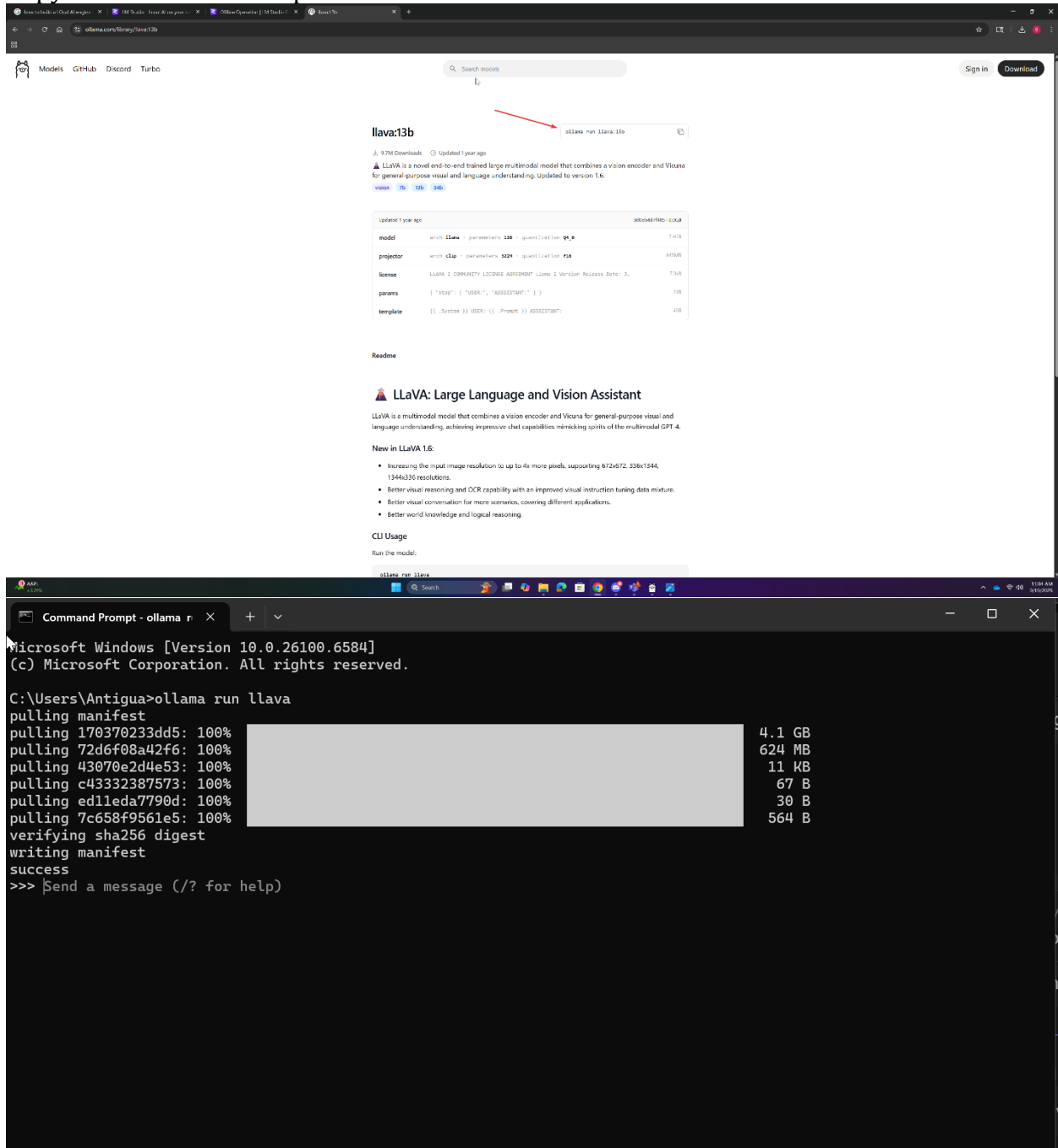
To actually import the Ai model that you want you need to go to ollama docs:



You can choose any Ai model that you want, for this lab I chose Llava:13b.



Copy the command on top and run it in the CMD to install the model



The screenshot shows the Ollama website interface and a Windows Command Prompt window. The website displays the 'llava:13b' model, which is a novel end-to-end trained large multimodal model that combines a vision encoder and Vicuna for general-purpose visual and language understanding. A red arrow points to the 'ollama run llava:13b' command. Below the model information, there is a table with details about the model, including its architecture, parameters, and license. The Command Prompt window shows the execution of the 'ollama run llava' command, which pulls the model from the Ollama library. The output shows the progress of pulling the model, including the size of the manifest and the progress of pulling the model data (4.1 GB).

llava:13b

3.7M Downloads · Updated 1 year ago

LLaVA is a novel end-to-end trained large multimodal model that combines a vision encoder and Vicuna for general-purpose visual and language understanding. Updated to version 1.6.

visit: [76](#) · [136](#) · [346](#)

Updated 1 year ago				xb204d460-6064
model	arch: llama	parameters: 13B	quantization: q4_0	7.67B
projector	arch: clip	parameters: 320M	quantization: f16	645MB
license	LLaMA 2 COMMUNITY LICENSE AGREEMENT Llama 2 Version Release Date: 3...			
params	{ "vocab": { "USER_TOKEN": "ASSISTANT_TOKEN" } }			
template	{ { System } } USER: { { Prompt } } ASSISTANT:			

Readme

LLaVA: Large Language and Vision Assistant

LLaVA is a multimodal model that combines a vision encoder and Vicuna for general-purpose visual and language understanding, achieving impressive chat capabilities mimicking spirits of the multimodal GPT-4.

New in LLaVA 1.6:

- Increasing the input image resolution to up to 4x more pixels, supporting 872x572, 336x1544, 1344x336 resolutions.
- Better visual reasoning and OCR capability with an improved visual instruction tuning data mixture.
- Better visual conversation for more scenarios, covering different applications.
- Better world knowledge and logical reasoning.

CLI Usage

Run the model:

```
ollama run llava
```

Microsoft Windows [Version 10.0.26100.6584]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Antigua>ollama run llava

pulling manifest

pulling 170370233dd5: 100% 4.1 GB

pulling 72d6f08a42f6: 100% 624 MB

pulling 43070e2d4e53: 100% 11 KB

pulling c43332387573: 100% 67 B

pulling ed11eda7790d: 100% 30 B

pulling 7c658f9561e5: 100% 564 B

verifying sha256 digest

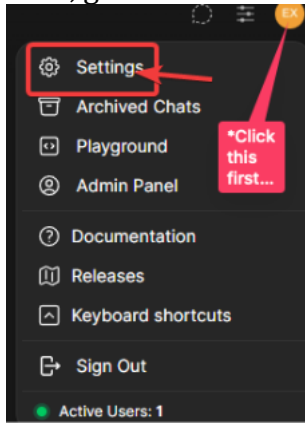
writing manifest

success

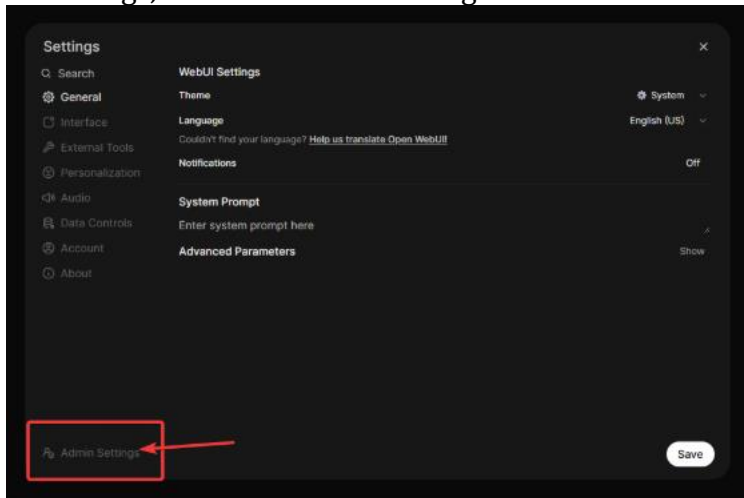
>>> Send a message (/? for help)



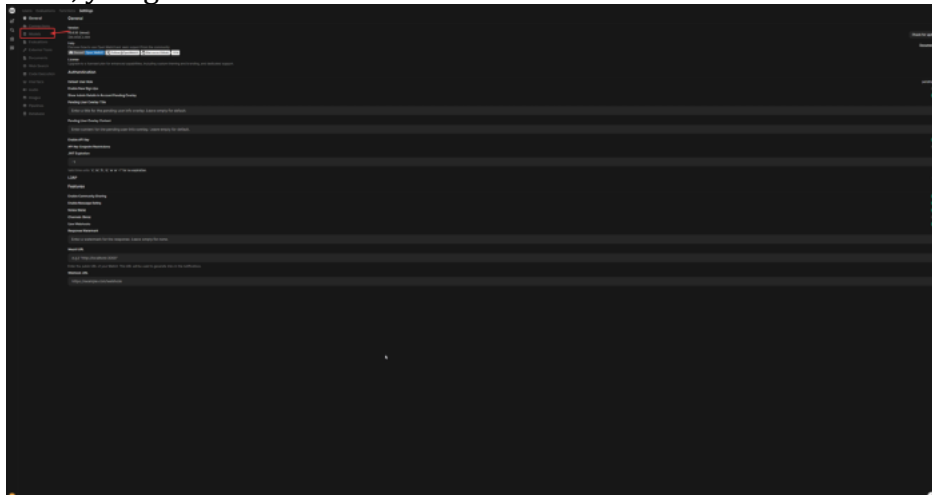
Now, go back to the WebUI to import the model.



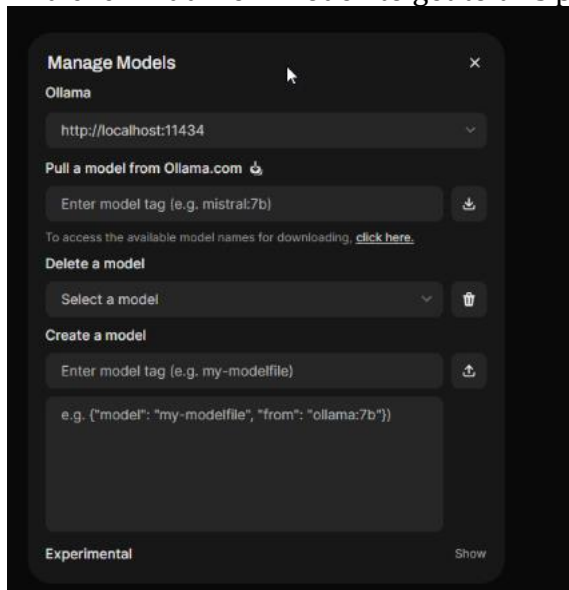
In Settings, click on 'Admin Settings':



Here, you go into the 'Models' tab:

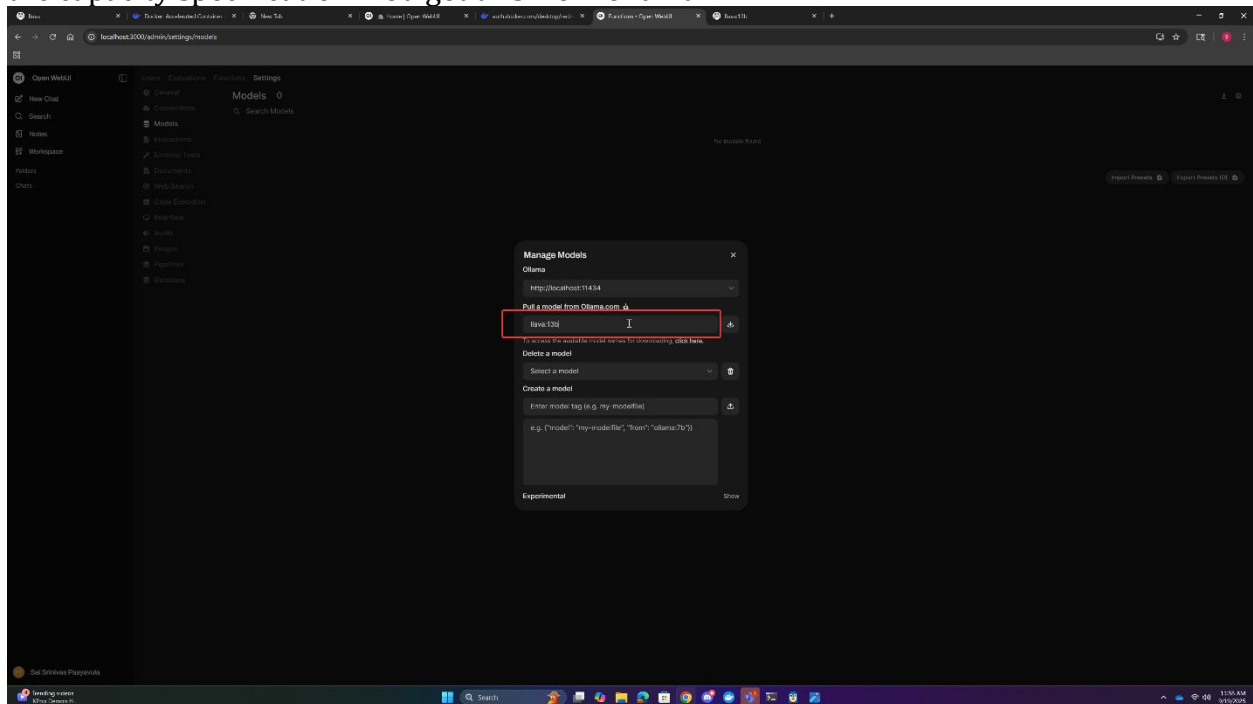


And click 'Add New Model' to get to this page:

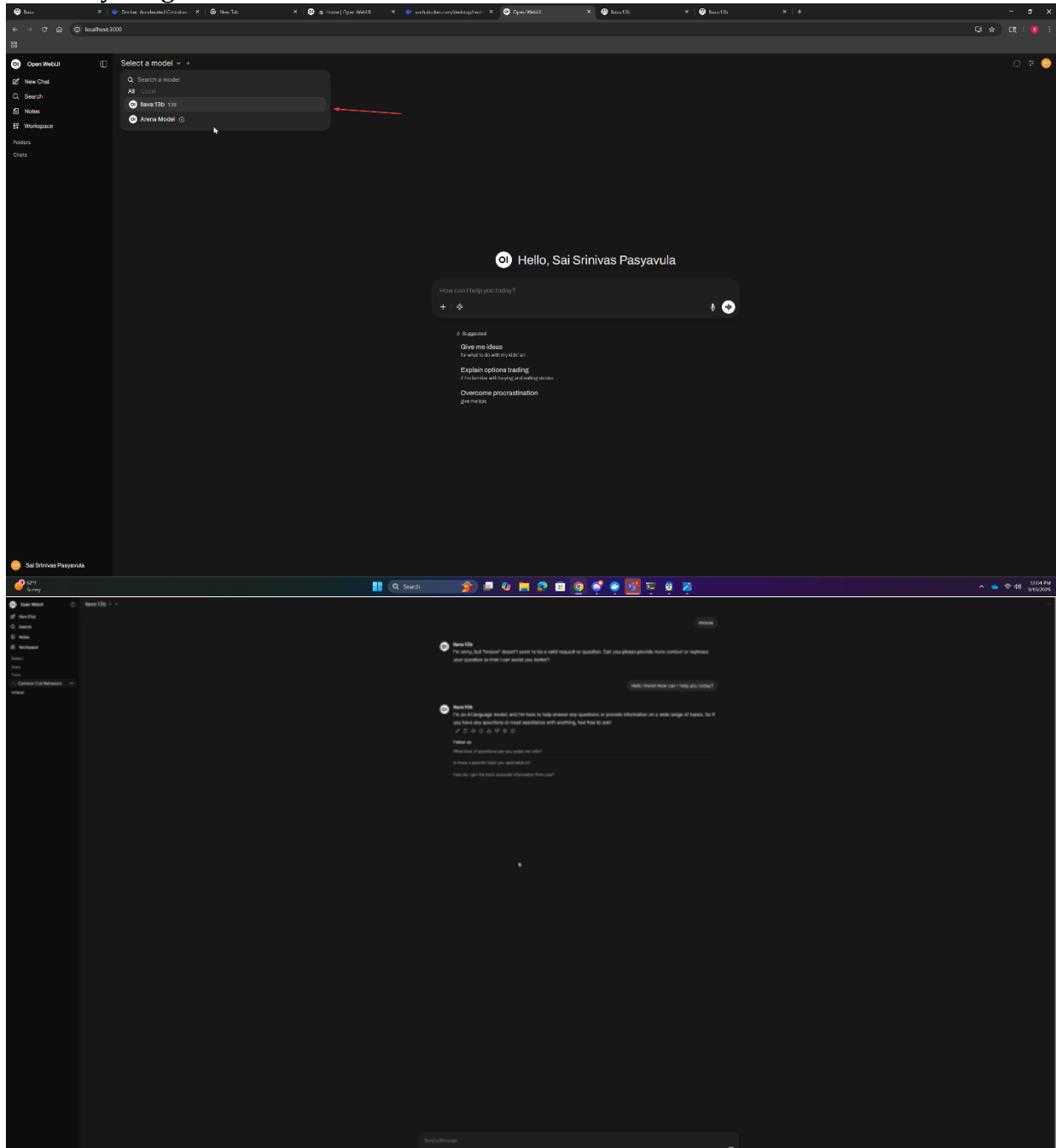


The 'Manage Models' dialog box is shown with a dark theme. It has a title bar with a close button. Below the title, there's a dropdown menu showing 'http://localhost:11434'. The main section is titled 'Pull a model from Ollama.com' and contains a text input field with the placeholder 'Enter model tag (e.g. mistral:7b)' and a download icon. Below this is a link: 'To access the available model names for downloading, [click here](#).' There are two more sections: 'Delete a model' with a 'Select a model' dropdown and a trash icon, and 'Create a model' with a text input field and a download icon. The 'Create a model' section has a text area with the example: 'e.g. {"model": "my-modelfile", "from": "ollama:7b"}'. At the bottom, there's a 'Show' button and the word 'Experimental'.

Please note that you'll need to type in the exact model's name for the AI models, including the capacity specification. You get this from ollama



After, it should appear in the model list that you can select to run on the WebUI and you can ask it anything.



Time to deploy it onto the network! It's time to go back into Command Prompt, and type in `ipconfig`:

```
Command Prompt
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 10:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Link-Local IPv6 Address . . . . : fe80::a7a:c8d5:8f1:9d74%14
IPv4 Address. . . . . : 192.168.40.86
Subnet Mask . . . . . : 255.255.254.0
Default Gateway . . . . . : 192.168.40.1

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

Ethernet adapter vEthernet (WSL (Hyper-V firewall)):

Connection-specific DNS Suffix . :
Link-Local IPv6 Address . . . . : fe80::996e:9aaf:9150:e3e8%56
IPv4 Address. . . . . : 172.25.144.1
Subnet Mask . . . . . : 255.255.240.0
Default Gateway . . . . . :

C:\Users\efy1228>
```

Use ipconfig for this

Here, you find your Wireless LAN adapter Wi-Fi, and the IPv4 address (These are private addresses, so nothing compromised here at all—tonnes of people have this same address in different regions).

Now, using this address, type it into your web browser's search bar like this: `192.168.40.46:3000`

Now, you see that `:3000` behind there. Well, that's the port number that the setup is on. log into your account credentials, and then you can do another prompt test. If it works, then you're all set!

For further testing, go onto the network on your phone or another device, and confirm you're on the same network (`192.168.40.0/23` in my case). Repeat the things above, and test if it would prompt the test.

Congratulations! You have officially finished setting up a small AI server that runs on the local network! If you have a better PC that has a great GPU in it, you can run more advanced models that have better accuracy.

Problems

I had problems with testing the AI network from another device as my phone. The computer had to be on the same network as the phone. The phone is automatically set to be automatically assigned, so we have to go in and manually set the Ip address on the phone. My phone, for some reason didn't let me do that for some reason. I believe it was because of



some parental control that my parents have on the ip addressing on my phone, so I had to use another device to make sure that the AI works well.

Another problem I ran into was some small issues regarding the actual installation of the AI but really minor that is easily fixable.

Conclusion

This lab served as a great introduction to Local AIs and how they are created. I learned the Basics of how to create my own local AI through two different ways. The local AIs are useful in today's world because it helps with cybersecurity, as no one can access the information on your local system.

